

Isobaric Vapor–Liquid Equilibrium of Binary Systems (Isopropyl Acetate/Isopropyl Alcohol + Dibutyl Ether/ Anisole) at 101.3 kPa

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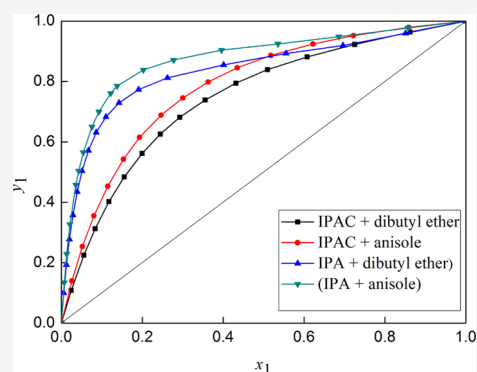


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ABSTRACT: Isopropyl acetate and isopropyl alcohol were azeotropic mixtures with a minimum boiling point. To break the azeotrope of the above binary system and separate it, dibutyl ether and anisole were chosen as the extractive solvents in this corresponding extractive distillation process. Therefore, the vapor–liquid equilibrium (VLE) measurements for (isopropyl alcohol + dibutyl ether), (isopropyl alcohol + anisole), (isopropyl acetate + dibutyl ether), and (isopropyl acetate + anisole) at 101.3 kPa were achieved by a modified Rose type recirculating still. The VLE data were checked by the van Ness test and pure component consistency test. The NRTL, UNIQUAC, and Wilson models were applied to correlate the determined VLE data. Furthermore, the suitability of entrainers was investigated by relative volatility and the effect of selected entrainers on the azeotrope. After that, the obtained binary interaction parameters can provide a basis for the simulation and optimization of the extractive distillation process.



1. INTRODUCTION

Isopropyl acetate (IPAC) is an essential solvent which has excellent solubility for lots of synthetic resins, for instance, ethyl cellulose, polystyrene, nitrocellulose, and methacrylate resins.¹ Further, it has a wide range of applications in pharmaceutical production as a dehydrating agent and an extractant.^{2,3} Generally, IPAC can be obtained by the esterification of acetic acid and isopropyl alcohol (IPA), while IPAC and IPA can form azeotrope, thus the final product contains IPA inevitably. Moreover, as a fine chemical and good solvent, IPA has widely applications in many aspects.^{4–6} Therefore, the separation of IPAC and IPA is of great significance. However, IPAC and IPA is a homogeneous azeotrope and its minimum boiling point is 353.59 K at atmospheric pressure.⁷ Thus, it is difficult to separate the azeotrope by conventional distillation.

Usually, some special distillations,⁸ for instance, extractive distillation,^{9–11} azeotropic distillation,¹² pressure-swing distillation,¹³ and salt distillation¹⁴ can be applied to separate azeotropic mixtures. After consulting the literatures of the studied azeotrope, few containing the azeotrope of (IPAC + IPA) were found. Casimiro et al.⁷ reported the vapor–liquid equilibrium (VLE) measurement for IPAC and IPA by a dynamic recirculating still, obtaining the minimum boiling point of the azeotrope at 101.32 kPa. Meanwhile, Andreatta et al. measured the VLE data and physical properties of the mixtures composed of IPAC, IPA, and the selected ILs at room temperature and atmospheric pressure, the adopted ILs, $[C_8mim][NTf_2]$, and $[C_4mim][NTf_2]$,^{15,16} break the azeotrope boiling point and can act as good solvents for the

extractive distillation process. Besides, as a low viscosity and cost solvent, dimethyl sulfoxide (DMSO) was chosen by Ding et al.¹⁷ to separate the azeotrope of IPAC and IPA at 101.3 kPa. After the VLE measurement of the related systems, the influence of DMSO was explored, which verified that it can be act as a promising extractant. Generally, extractive distillation^{18–21} is widely recognized to separate the azeotropic systems. According to the selection criteria of the extractive solvent proposed by Gmehling and Möllmann,²² dibutyl ether and anisole were selected as the extractive solvent candidates. After searching the NIST²³ and DDB²⁴ databases, the VLE data of (IPAC + dibutyl ether), (IPAC + anisole), (IPA + dibutyl ether), and (IPA + anisole) at 101.3 kPa have not been reported in previous literatures. Therefore, to separate the azeotrope (IPAC + IPA) mixture and obtain the parameters to design and simulate the extractive distillation process,²⁵ the above-mentioned VLE data are required.

In this work, dibutyl ether and anisole, as the entrainers, were selected to break the azeotropic binary system of (IPAC + IPA). Therefore, the measurement for these four binary systems, (IPAC + dibutyl ether), (IPAC + anisole), (IPA + dibutyl ether), and (IPA + anisole) was done at 101.3 kPa.

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Table 1. CAS Number, Suppliers, Purity and Boiling Temperature (T_b) of the Chemical Reagents

component	CAS	suppliers	mass fraction	water content (wt %)	T_b/K^b		analysis method
					exp.	lit.	
IPAC	108-21-4	Shandong Xiya Chemical Co., Ltd.	≥ 0.990	0.03%	361.67	361.65 ³¹ 361.65 ³² 361.45 ¹⁷	GC ^a
IPA	67-63-0	Chengdu Kelong Chemical Co., Ltd.	≥ 0.997	0.02%	355.20	355.37 ¹⁷ 355.33 ³³ 355.35 ³⁴	GC ^a
dibutyl ether	142-96-1	Chengdu Kelong Chemical Co., Ltd.	≥ 0.990	0.01%	414.00	414.55 ³⁵ 414.66 ³⁶ 413.44 ³⁷	GC ^a
anisole	100-66-3	Tianjin Kemiou Chemical Reagent Co., Ltd.	≥ 0.990	0.01%	426.98	426.75 ³⁷ 426.54 ³⁸ 426.54 ³⁹	GC ^a

^aGas chromatograph. ^bThe experimental pressure is 101.3 kPa, the standard uncertainties u of P and T are $u(P) = 0.35$ kPa, $u(T) = 0.35$ K.

Moreover, the van Ness test²⁶ and pure component consistency test²⁷ were used to verify the consistency for the measured data. Besides, the experimental VLE data were correlated by the nonrandom two-liquid (NRTL),²⁸ universal quasichemical (UNIQUAC),²⁹ and Wilson³⁰ thermodynamic models, then the binary interaction parameters of these four binary systems were obtained. Furthermore, by comparing the relative volatility and the phase behavior influence of the selected entrainers on the azeotropic system, the suitability and optimum dosage of entrainers were explored.

2. EXPERIMENTAL SECTION

2.1. Chemicals and Materials. Analytical chemicals IPAC, IPA, dibutyl ether and anisole were purchased commercially. The mass fractions of the above chemicals were tested by gas chromatography (GC). The water contents were measured by a Karl Fischer Moisture Meter (KLS701). All of them were used without further treatment. The CAS numbers, suppliers, mass fractions, and water contents of the above chemicals are listed in Table 1. Besides, the measured boiling temperatures of the above four chemicals were compared with the literature data in Table 1.

2.2. Apparatus and Procedures. The experimental VLE measurements for (IPAC + dibutyl ether), (IPAC + anisole), (IPA + dibutyl ether) and (IPA + anisole) were determined at 101.3 kPa with a modified Rose type recirculating still.⁴⁰ The temperature was checked by an accurate mercury thermometer, and the pressure was determined by a mercury U-shaped manometer. To keep the pressure constant, a pressure control system was applied in this work. As shown in Figure 1, the vacuum pump was connected to the equilibrium still, and the needle valve was adjusted slowly to ensure that the pressure could be kept at 101.3 kPa during the experiment. About 50 min equilibrium, until the fluctuation of temperature was within 0.1 K,⁴¹ the VLE state was then completed. After that, the vapor and liquid phase samples were withdrawn by syringe contemporaneously, and then analyzed by GC immediately. The related measurements could also be found in the previous literatures.^{42–45}

2.3. Analysis. The GC (Lunan Ruihong Chemical Instruments Co., Ltd. SP6890) was used to analyze the vapor and liquid samples, which was equipped with a Porapak Q (3 mm $\times$ 2 m). The temperatures, carrier gas and other analysis conditions for GC are listed in Table 2. Besides, the area calibration normalization method was used to calibrate

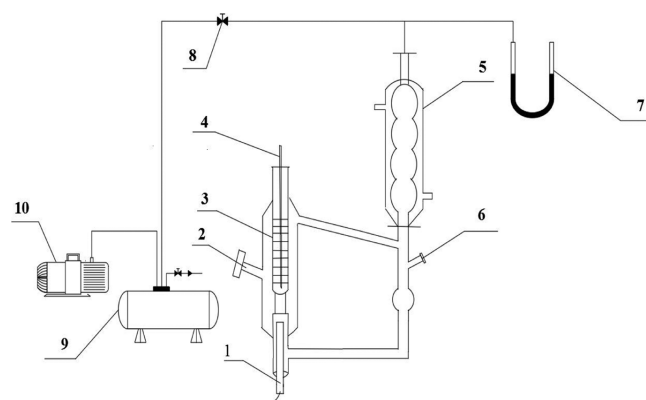


Figure 1. VLE apparatus: (1), heating rod; (2), liquid phase sample port; (3), thermometer sleeve tube; (4), mercury thermometer; (5), condenser; (6), vapor phase sample port, (7), mercury U-shaped manometer, (8), needle valve, (9), buffer tank, and (10), vacuum pump.

Table 2. Analysis Conditions for the Gas Chromatography

name	characteristic	description
column	type	packing column
	specification	Porapak Q (3 mm $\times$ 2 m)
carrier gas	type	hydrogen
	pressure	0.17 MPa
injector	temperature	453.15 K
column	temperature	393.15 K
detector	type	thermal conductivity detector
	temperature	463.15 K

the actual compositions.⁴⁶ For the sample analysis, the peak area of GC was calibrated by five different standard samples that covered the whole composition range. Then, all of the samples were measured at least three times to make sure the deviations less than 0.001, the average value was adopted.⁴⁷

3. RESULTS AND DISCUSSION

3.1. Experimental Data. The VLE measurements for (IPAC + dibutyl ether), (IPAC + anisole), (IPA + dibutyl ether) and (IPA + anisole) four binary systems were done at 101.3 kPa, and all the measured results are listed in Tables 3–6. Moreover, the calculated activity coefficients (γ) and excess Gibbs energy (G^E) of the studied systems are all listed

Table 3. Values of Isobaric VLE Measurement, Activity Coefficient (γ) and Excess Gibbs Energy (G^E) for the System of IPAC (1) + Dibutyl Ether (2) at 101.3 kPa^a

T/K	x_1	y_1	γ_1	γ_2	G^E
361.67	1.0000	1.0000			0
364.93	0.8623	0.9641	1.0087	1.2682	122.01
368.49	0.7253	0.9228	1.0283	1.1977	213.87
372.02	0.6076	0.8816	1.0541	1.1312	248.65
375.51	0.5095	0.8394	1.0796	1.0846	246.15
378.79	0.4313	0.7944	1.0976	1.0688	245.65
382.52	0.3555	0.7389	1.1144	1.0554	232.90
386.14	0.2926	0.6817	1.1300	1.0397	203.23
389.32	0.2443	0.6258	1.1397	1.0321	180.73
392.62	0.1998	0.5620	1.1462	1.0274	159.62
396.34	0.1554	0.4840	1.1519	1.0216	131.84
399.91	0.1175	0.4024	1.1564	1.0160	103.24
403.50	0.0833	0.3125	1.1581	1.0114	75.75
406.70	0.0554	0.2253	1.1607	1.0076	52.19
410.63	0.0242	0.1083	1.1624	1.0038	25.12
414.00	0.0000	0.0000			0

^aStandard uncertainties u of T , P , x and y are $u(T) = 0.35$ K, $u(P) = 0.35$ kPa, $u(x) = 0.0072$, $u(y) = 0.0075$.

Table 4. Values of Isobaric VLE Measurement, Activity Coefficient (γ) and Excess Gibbs Energy (G^E) for the System of IPAC (1) + Anisole (2) at 101.3 kPa^a

T/K	x_1	y_1	γ_1	γ_2	G^E
361.67	1.0000	1.0000			0
365.76	0.8587	0.9793	1.0027	1.0595	31.76
370.42	0.7221	0.9519	1.0048	1.0474	50.27
374.36	0.6220	0.9245	1.0074	1.0438	64.67
379.07	0.5178	0.8865	1.0121	1.0373	75.25
383.46	0.4350	0.8456	1.0153	1.0320	77.71
387.79	0.3627	0.7983	1.0205	1.0306	85.68
392.13	0.3003	0.7454	1.0247	1.0253	80.90
396.39	0.2465	0.6888	1.0321	1.0135	58.97
401.15	0.1932	0.6151	1.0420	1.0072	45.94
405.26	0.1533	0.5423	1.0457	1.0057	39.40
409.73	0.1145	0.4532	1.0503	1.0047	33.14
414.10	0.0802	0.3553	1.0606	1.0035	27.36
418.22	0.0518	0.2536	1.0661	1.0019	17.95
422.40	0.0259	0.1399	1.0709	1.0002	6.92
426.98	0.0000	0.0000			0

^aStandard uncertainties u of T , P , x and y are $u(T) = 0.35$ K, $u(P) = 0.35$ kPa, $u(x) = 0.0072$, $u(y) = 0.0075$.

in Tables 3–6 as well. The corresponding T – x – y phase charts are plotted in Figure 2. After consulting the literatures about the studied azeotrope, we found that the VLE measurements for (IPAC + anisole) have been studied by Bhoi et al.⁴⁸ at 93.9 kPa, and the measurement for (IPA + anisole) have been studied by Mali et al.⁴⁹ at 96.15 kPa. For comparison, the isobaric VLE data for (IPAC + anisole) and (IPA + anisole) measured at 101.3 kPa in this work and reported in the literatures mentioned above are shown in Figure 2b,d, respectively. The difference between them may be caused by the different apparatus and pressure.

3.2. VLE Calculation. The VLE relationship is expressed as follows⁵⁰

Table 5. Values of Isobaric VLE Measurement, Activity Coefficient (γ) and Excess Gibbs Energy (G^E) for the System of IPA (1) + Dibutyl Ether (2) at 101.3 kPa^a

T/K	x_1	y_1	γ_1	γ_2	G^E
355.20	1.0000	1.0000			0
356.89	0.8509	0.9597	1.0547	1.7940	392.99
358.62	0.6963	0.9195	1.1540	1.6432	747.02
360.07	0.5557	0.8929	1.3275	1.4121	930.18
363.11	0.4010	0.8553	1.5695	1.2589	961.99
367.14	0.2622	0.8120	1.9620	1.1414	837.21
371.01	0.1908	0.7732	2.2325	1.0897	687.10
374.91	0.1426	0.7296	2.4578	1.0670	572.93
378.89	0.1101	0.6832	2.6017	1.0488	465.28
382.82	0.0861	0.6315	2.6984	1.0399	385.83
386.92	0.0667	0.5713	2.7596	1.0347	320.19
391.14	0.0510	0.5042	2.7893	1.0275	253.98
394.98	0.0390	0.4347	2.7959	1.0258	211.93
398.93	0.0284	0.3580	2.8105	1.0211	164.53
402.75	0.0197	0.2778	2.8134	1.0157	119.28
406.42	0.0123	0.1932	2.8239	1.0117	82.05
410.24	0.0055	0.1001	2.9435	1.0050	37.12
414.00	0.0000	0.0000			0

^aStandard uncertainties u of T , P , x and y are $u(T) = 0.35$ K, $u(P) = 0.35$ kPa, $u(x) = 0.0075$, $u(y) = 0.0087$.

Table 6. Values of Isobaric VLE Measurement, Activity Coefficient (γ) and Excess Gibbs Energy (G^E) for the System of IPA (1) + Anisole (2) at 101.3 kPa^a

T/K	x_1	y_1	γ_1	γ_2	G^E
355.20	1.0000	1.0000			0
357.11	0.8575	0.9772	1.0565	1.6355	347.96
359.89	0.6858	0.9487	1.1508	1.4900	663.19
362.19	0.5353	0.9245	1.3158	1.3521	864.53
364.98	0.3960	0.9039	1.5661	1.1862	852.02
369.11	0.2773	0.8712	1.8529	1.1333	802.29
373.99	0.2028	0.8386	2.0510	1.0725	626.54
380.25	0.1380	0.7855	2.2800	1.0516	496.63
382.90	0.1218	0.7606	2.2914	1.0497	457.06
388.78	0.0927	0.6998	2.2929	1.0421	369.57
393.02	0.0754	0.6500	2.2953	1.0360	311.53
399.32	0.0540	0.5648	2.3053	1.0286	238.26
403.30	0.0429	0.5040	2.3073	1.0237	195.50
406.02	0.0357	0.4573	2.3289	1.0232	176.57
413.10	0.0210	0.3269	2.3298	1.0133	105.43
417.67	0.0130	0.2290	2.3359	1.0099	71.93
421.72	0.0068	0.1344	2.3612	1.0060	41.16
426.98	0.0000	0.0000			0

^aStandard uncertainties u of T , P , x and y are $u(T) = 0.35$ K, $u(P) = 0.35$ kPa, $u(x) = 0.0085$, $u(y) = 0.0096$.

$$P y_i \phi_i = x_i \gamma_i \phi_i^s P_i^s \exp \left[\frac{V_i^L (P - P_i^s)}{RT} \right] \quad (1)$$

Because the pressure was 101.3 kPa, the Poynting factor, ϕ_i^s and ϕ_i associated with nonideality were all close to 1. Equation 1 can be simplified and calculated by the following equations

$$\gamma_i = \frac{P y_i}{P_i^s x_i} \quad (2)$$

where x_i and y_i represent the equilibrium mole fraction of component i in the liquid phase and the vapor phase; P is the

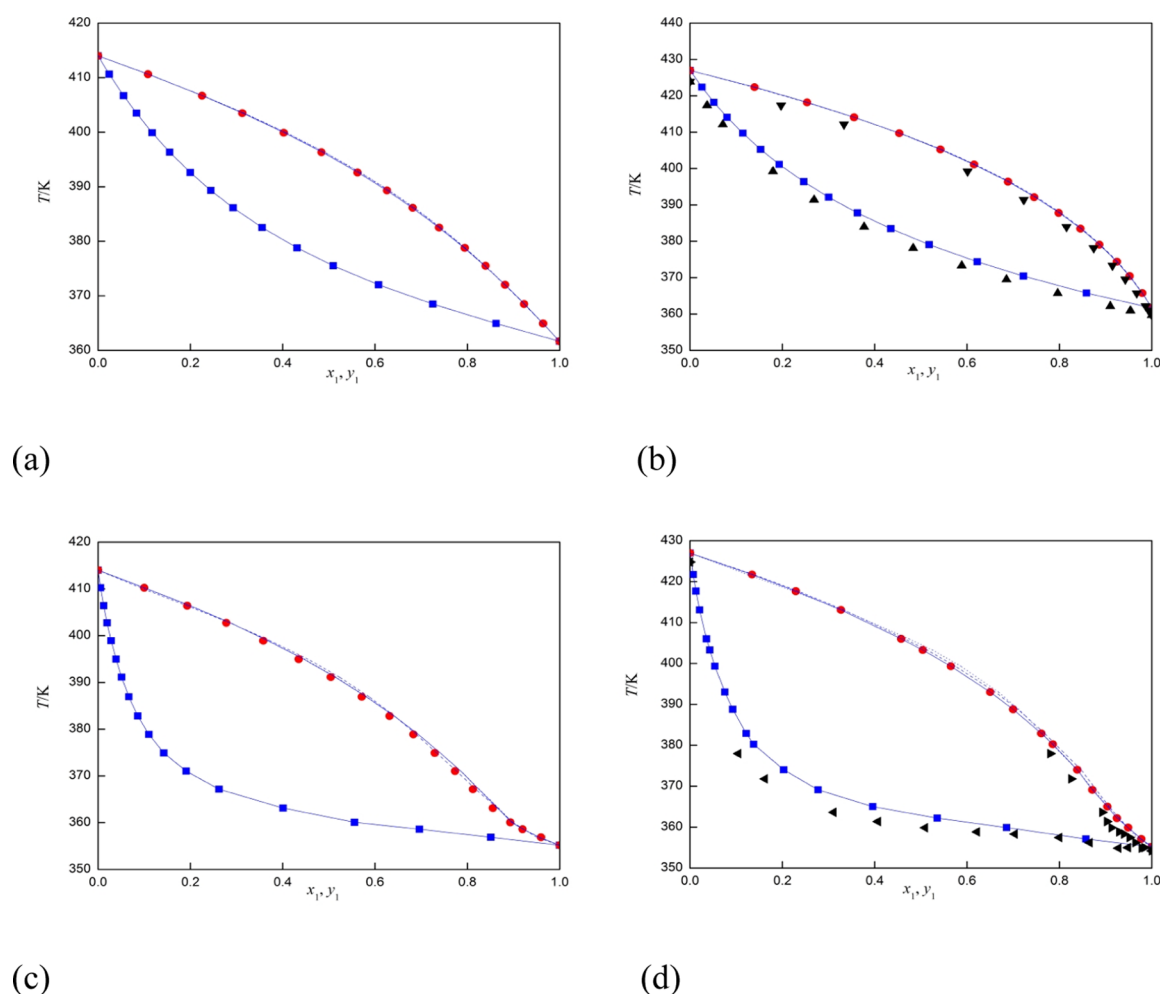


Figure 2. T - x - y diagram for the four binary systems at 101.3 kPa: (a), IPAC (1) + dibutyl ether (2); (b), IPAC (1) + anisole (2); (c), IPA (1) + dibutyl ether (2); (d), IPA (1) + anisole (2); blue ■, T - x , experimental values; red ●, T - y , experimental values; ▲, T - x , literature values;⁴⁸ ▼, T - y , literature values;⁴⁸ ◀, T - x , literature values;⁴⁹ ▶, T - y , literature values;⁴⁹ blue —, calculated by the NRTL model; blue ---, calculated by the UNIQUAC model; blue ···, calculated by the Wilson model.

Table 7. Parameters for Antoine Equation^a

component	C_{1i}	C_{2i}	C_{3i}	C_{4i}	C_{5i}	C_{6i}	C_{7i}	C_{8i}/K	C_{9i}/K
IPAC	42.85	-5563.90	0	0	-3.88	2.48×10^{-18}	6	199.75	532.00
IPA	103.81	-9040.00	0	0	-12.68	5.54×10^{-6}	2	185.26	508.30
dibutyl ether	65.32	-7537.60	0	0	-7.06	9.14×10^{-18}	6	175.30	584.10
anisole	121.15	-9307.70	0	0	-16.69	1.49×10^{-2}	1	235.65	645.60

^aTaken from Aspen property databank.

pressure of the four systems and P_i^s delegates the saturation pressure of pure component i , which was calculated by the extended Antoine expression as follows

$$\ln(P_i^s) = C_{1i} + \frac{C_{2i}}{T + C_{3i}} + C_{4i}T + C_{5i} \ln T + C_{6i}T^{C_{7i}} \quad (3)$$

$$C_{8i} \leq T \leq C_{9i}$$

The constants for the used four chemicals were acquired from the Aspen databank,⁵¹ which are listed in Table 7.

To estimate the nonideality of these binary systems, values of G^E are calculated by eq 4

$$G^E = RT(x_1 \ln \gamma_1 + x_2 \ln \gamma_2) \quad (4)$$

Values of γ_i and G^E were achieved based on the NRTL model and are expressed in Figure 3. Obviously, positive deviations from Raoult's law are shown for the four systems, and the results abide by the order (IPA + dibutyl ether) > (IPA + anisole) > (IPAC + dibutyl ether) > (IPAC + anisole).

3.3. Thermodynamic Consistency Test. In this work, the van Ness test^{26,52} and pure component consistency test^{27,53} were applied to verify the thermodynamic consistency of the experimental data for the four binary systems.

The van Ness test is defined as follows

$$\Delta y = \frac{1}{N} \sum_{i=1}^N 100 |y_i^{\text{exp}} - y_i^{\text{cal}}| \quad (5)$$

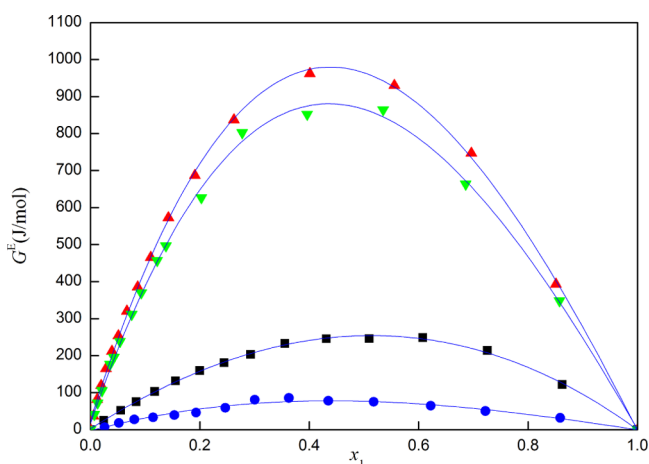


Figure 3. Experimental G^E for the four binary systems and the calculated results by the NRTL model vs x_1 : ■, IPAC (1) + dibutyl ether (2); blue ●, IPAC (1) + anisole (2); brown ▲, IPA (1) + dibutyl ether (2); green ▼, IPA (1) + anisole (2); solid line represents the trend line.

$$\Delta P = \frac{1}{N} \sum_{i=1}^N 100 \left| \frac{P_i^{\text{exp}} - P_i^{\text{cal}}}{P_i^{\text{exp}}} \right| \quad (6)$$

where N is the number of experimental point; y_i^{exp} refers to the experimental data of mole fractions in vapor phase; P_i^{exp} refers to experimental pressure; while y_i^{cal} , P_i^{cal} are achieved based on the NRTL model. The values of ΔP and Δy checked by the van Ness test are supposed to be less than 1 and summarized in Table 8. As shown in Table 8, all the calculated values of ΔP and Δy meet this requirement, indicating that the experimental VLE data passed the van Ness test.

Table 8. Results of Thermodynamic Consistency Checked by van Ness Test

system	ΔP	Δy
IPAC + dibutyl ether	0.0216	0.5472
IPAC + anisole	0.0088	0.1486
IPA + dibutyl ether	0.0754	0.7521
IPA + anisole	0.1133	0.6933

The pure component consistency test is presented as follows

$$\Delta P_1^0 = \left| \frac{P_{\text{bubble}}(x_1 \rightarrow 1) - P_1^0}{P_1^0} \right| \quad (7)$$

$$\Delta P_2^0 = \left| \frac{P_{\text{bubble}}(x_1 \rightarrow 0) - P_2^0}{P_2^0} \right| \quad (8)$$

where P_1^0 and P_2^0 are the vapor pressures of the pure component and P_{bubble} is the bubble point pressure of the mixture. The values of ΔP_1^0 and ΔP_2^0 are expected to be less than 1. For the four binary systems, the calculated values of ΔP_1^0 and ΔP_2^0 are listed in Table 9. As shown in Table 9, all the calculated results are less than 1, indicating that the experimental data passed the pure component consistency test.

3.4. Correlation. The measured results for the four binary systems of (IPAC + dibutyl ether), (IPAC + anisole), (IPA + dibutyl ether), and (IPA + anisole) were correlated using the NRTL,²⁸ UNIQUAC,²⁹ and Wilson³⁰ models. In the

Table 9. Pure Component Consistency Test for Thermodynamic Consistency

system	ΔP_1^0	ΔP_2^0
IPAC (1) + dibutyl ether (2)	0.0027	0.0017
IPAC (1) + anisole (2)	0.0027	0.0078
IPA (1) + dibutyl ether (2)	0.0258	0.0017
IPA (1) + anisole (2)	0.0258	0.0078

Table 10. Structural Parameters for the UNIQUAC Model^a

component	r	q
IPAC	4.1523	3.6520
IPA	2.9137	2.5276
dibutyl ether	6.0593	5.1520
anisole	4.1668	3.2080

^aTaken from Aspen property databank.

UNIQUAC model, both the volume parameter r and area parameter q are listed in Table 10. The objective function is defined as follows

$$\text{OF} = \sum_i^N \left[\left(\frac{P_i^{\text{exp}} - P_i^{\text{cal}}}{\sigma_P} \right)^2 + \left(\frac{T_i^{\text{exp}} - T_i^{\text{cal}}}{\sigma_T} \right)^2 + \left(\frac{x_i^{\text{exp}} - x_i^{\text{cal}}}{\sigma_x} \right)^2 + \left(\frac{y_i^{\text{exp}} - y_i^{\text{cal}}}{\sigma_y} \right)^2 \right] \quad (9)$$

where N refers to the amount of data points; T refers to temperature of the binary systems; P refers to equilibrium pressure; x and y represent the liquid and vapor phase mole fraction of component i , respectively; σ is the standard deviation; the superscript “exp” stands for the experimental data; and the superscript “cal” represents the calculated values.

For comparison, the root-mean-square deviation (rmsd) was calculated to assess the difference between the calculated values and experimental one, which are presented in Table 11. The values of rmsd (y_i) and rmsd (T) are less than 0.0097 and 0.55 K. Therefore, the used models are appreciated to the VLE data of the four binary systems. The interaction parameters are regressed and also presented in Table 11.

4. SUITABILITY OF ENTRAINERS

4.1. Relative Volatility. Generally, the suitability of the solvents was examined by the relative volatility (α_{12}) defined as follows^{54,55}

$$\alpha_{12} = \frac{y_1/x_1}{y_2/x_2} \quad (10)$$

where x_i is the mole fraction of component i in the liquid phase and y_i is the vapor phase. The relative volatility of mixtures IPA (1) + IPAC (2) with the entrainers (dibutyl ether and anisole) are calculated by the binary parameters of NRTL obtained from Table 11. Comparing with the relative volatility of azeotrope IPA (1) + IPAC (2) without an entrainer,⁷ the mixtures of IPA (1) + IPAC (2) with the entrainers (dibutyl ether and anisole) are presented in Figure 4. As shown in Figure 4, the relative volatility values for the mixture IPA (1) +

Table 11. Correlation Parameters, Standard Deviations of Vapor Mole Fraction (y_1) and Boiling Temperature (T) for the Studied Four Systems at 101.3 kPa

model	parameters				rmsd	
	a_{ij}	a_{ji}	b_{ij}/K	b_{ji}/K	y_1^a	T/K^b
IPAC + dibutyl ether						
NRTL ^c	0.0103	−0.0071	332.72	−180.86	0.0063	0.09
UNIQUAC ^d	−0.0057	0.0054	−37.90	8.78	0.0059	0.08
Wilson ^e	0.0055	−0.0083	102.02	−242.53	0.0062	0.09
IPAC + anisole						
NRTL	−0.0019	0.0023	−145.02	192.57	0.0019	0.04
UNIQUAC	0.0002	−0.0001	−45.74	52.67	0.0022	0.03
Wilson	−0.0015	0.0012	−143.82	99.11	0.0019	0.04
IPA + dibutyl ether						
NRTL	0.0064	−0.0050	255.09	245.27	0.0090	0.33
UNIQUAC	−0.0012	0.0018	112.65	−341.76	0.0087	0.33
Wilson	0.0041	−0.0066	−245.07	−307.11	0.0091	0.34
IPA + anisole						
NRTL	0.0141	−0.0095	357.19	110.63	0.0094	0.51
UNIQUAC	−0.0058	0.0057	−45.12	−104.13	0.0090	0.48
Wilson	0.0070	−0.0130	−145.13	−367.98	0.0097	0.55

$^a \text{rmsd}(y_1) = \sqrt{\sum_{i=1}^N \frac{(y_{1,i}^{\text{exp}} - y_{1,i}^{\text{cal}})^2}{N}}$. $^b \text{rmsd}(T) = \sqrt{\sum_{i=1}^N \frac{(T_i^{\text{exp}} - T_i^{\text{cal}})^2}{N}}$. $^c \text{NRTL}$, $\tau_{ij} = a_{ij} + b_{ij}/T$ (the value of α was fixed at 0.3). $^d \text{UNIQUAC}$, $\tau_{ij} = \exp(a_{ij} + b_{ij}/T)$. $^e \text{Wilson}$, $\ln A_{ij} = a_{ij} + b_{ij}/T$.

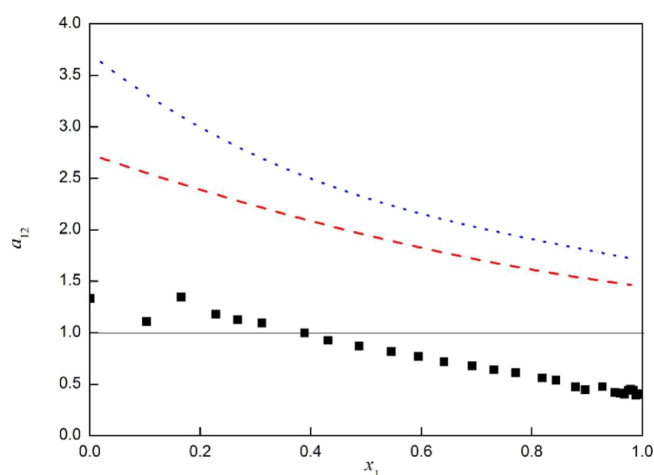


Figure 4. Relative volatility vs x_1 plot: ■, literature values for IPA (1) + IPAC (2) without entrainer; red --- and blue ···, calculated by the NRTL model with the entrainers dibutyl ether and anisole.

IPAC (2) with the added entrainers are all greater than 1, showing that both dibutyl ether and anisole can break the azeotrope. Besides, the calculated values of relative volatility with anisole show larger deviations from the unity than those with dibutyl ether, indicating that anisole has better extraction effect.

4.2. Effect of Entrainers. According to the binary parameters of NRTL, the phase behavior influence of the dibutyl ether and anisole added into the azeotropic system (IPAC + IPA) are expressed in Figure 5. As seen in x – y plots, when the ratio of extractive solvents flowrate (S) to the feed flowrate (F) was 0.2:1, the azeotropic point of IPA (1) + IPAC (2) was not broken under the action of the two extractants. When the S/F increased to 0.3:1, the two extractive solvents just broke the azeotropic point of IPA (1) + IPAC(2). Therefore, the optimum S/F for breaking the azeotropic point of IPA (1) + IPAC(2) was 0.3:1. Evidently, the effect of

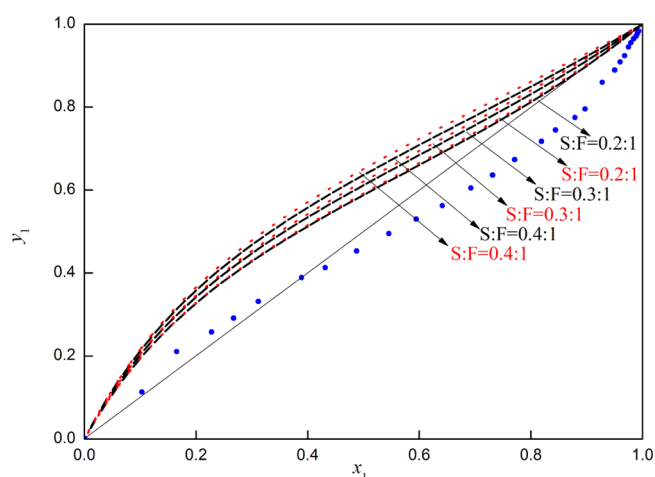


Figure 5. Influence on VLE data for the system of IPA (1) + IPAC (2) with different extractive solvents and different amounts of extractive solvent: blue ●, literature values⁷ without an entrainer; ----, calculated by the NRTL model with the extractive solvent dibutyl ether; red ···, calculated by the NRTL model with the extractive solvent anisole.

anisole is more obvious than dibutyl ether, which proves that anisole is a better choice to extract the azeotropic system as well.

5. CONCLUSIONS

In this work, to break the binary azeotrope (IPAC + IPA), dibutyl ether, and anisole were chosen as extractive solvents. The VLE measurements for the binary systems of (IPAC + dibutyl ether), (IPAC + anisole), (IPA + dibutyl ether), and (IPA + anisole) were determined at 101.3 kPa. Then, the van Ness test and pure component consistency test were applied to check the thermodynamic consistency of the measured data. Meanwhile, the experimental data were correlated by the NRTL, UNIQUAC, and Wilson models, and the binary

parameters were obtained. Besides, the values of $\text{rmsd}(y_1)$ and $\text{rmsd}(T)$ were less than 0.0097 and 0.55 K, respectively, which suggest that the measured data meet the regression requirements. Furthermore, by investigating the relative volatility and the phase behavior influence of the selected entrainers on the azeotropic system, it is suggested that anisole is a better choice to extract the azeotrope (IPAC + IPA) than dibutyl ether, and the optimum S/F for breaking the azeotropic point of the system was 0.3:1. The measured VLE data and the regressed parameters can provide a basis for the design and simulation of the distillation process.

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Notes

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